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Homework Set 3

~~5.8, 5.10, 5.16, 5.17, (5.22–5.24), 5.27, 5.31, 5.33, 5.34, 5.36, 5.37, 5.39~~

5.8) $P(D+) = 0.35\%$ $P(D-) = 99.65\%$ $P(T+|D+) = 99.9\%$

$P(T-|D-) = 99.9\%$ $P(T+|D-) = 0.01\%$

$$P(D+|T+) = \frac{P(T+|D+)\cdot P(D+)}{P(T+|D+)\cdot P(D+) + P(T+|D-)\cdot P(D-)} = \boxed{77.8}$$

5.10) 10 possible species. How many pair combinations?

$$\frac{10!}{(2!\cdot 8!)} = \frac{10\cdot 9}{2\cdot 1} = \frac{90}{2} = \boxed{45}$$ possible pairs

5.16 $P(D+) = 0.25$ $P(D-) = 0.75$ 6 children

no children have disease

$$\left(\frac{3}{4}\right)^6 = \frac{3}{4} \quad \frac{729}{4096} = 0.17797$$ round to 2 sig $\boxed{.18}$ no disease

one child has disease

$$\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^5 = \frac{243}{4096} = 0.059 = \boxed{0.06}$$ 1 has

0.02 0.02 2 have

0.01

$k=6$ $p = 3/4$ $q = 1/4$

5.17) $P = 1/68$ $q = 63/68$ $x = 4$ 1 dil $1/64$

a) $x=0$

$P(0) = (4!/(0!\cdot 4!))\cdot\left(\frac{1}{68}\right)^0\cdot\left(\frac{67}{68}\right)^4 = 0.942461$

$x=1$ $P(1) = (4!/(1!\cdot 3!))\cdot \frac{1}{68}\left(\frac{67}{68}\right)^3 = 0.05626$

$x=2$ $P(2) = (4!/(2!\cdot 2!))\cdot\left(\frac{1}{68}\right)^2\cdot\left(\frac{63}{68}\right)^2 = 0.00126$

$x=3$ $P(3) = (4!/(3!\cdot 1!))\cdot\left(\frac{1}{68}\right)^3\cdot\left(\frac{63}{68}\right)^1 = 0.0000125$

$x=4$ $P(4) = (4!/(4!\cdot 0!))\cdot\left(\frac{1}{68}\right)^4 = 4.67\times10^{-8}$

c) $p(x\geq 1) = 1 - p(0)$ $p(0) = 0.942461 = 5.7539\times10^{-2}$

5.22) 2 tails TT HT HH TH $\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$

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7.5, 7.4, 7.9, 7.11, 7.16, 7.21, 7.27

5.23) 3d6 2 twos & 1 four $\frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = 1/36$

5.24) p(rain Sat) = 90%

p(rain Sun) = .70%

$p(A \cup B) = p(A) + p(B) - P(A \cap B)$ where $P(A \cap B) = p(A) \cdot p(B)$

$(.9) + (.7) - (.63) = .97$ $P(A \cap B) = 63/100$

or $\boxed{97\%}$

5.27) 26 black 26 red

$\left(\frac{26}{52}\right) \cdot \left(\frac{26}{51}\right) = 0.254901961 = 25.49\% \cdot 2 = \boxed{50.98\%}$

5.31) 10 coins 6 heads $\binom{10}{6} = \frac{10!}{6!(10-6)!}$

$= \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = \boxed{210}$

5.33) 6 coins 3 heads exactly 0.3125

5.34) 5d6 exactly 3 twos = 0.03125

5.36) p(♀) = 0.5 6 kids

a) 2 or fewer girls = 0.34375

5.37) P(👑) = .1

a) None are 👑 = .07178 or 7.18%

b) ≥1 are 👑 = 1 - .07178 = 0.92821 or 92.82%

5.39) 5E 8 buc | 3E 8 vinu

$p(x) = \frac{\mu^x e^{-\mu}}{x!}$

mean = 3/5 = .6 = μ

$x = 3$

$\frac{.6^3 e^{-.6}}{3!} = 0.019757$ or 1.98%

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7.5)

	Migrators	Non Migrators	Total
♀	25 (12.84)	2 (14.16)	27
♂	+ 4 (16.16)	+ 30 (17.84)	34
	29	32	61 = N

$\chi^2 = 39.42 > CV = 3.84$

$df = (rows-1)\cdot(col-1) = 1$

reject H_0

7.8)

		Ill	Not Ill	Total	Atack Rate
Chic	Eat	37	5	42	88
	No Eat	11	11	22	50
Pot Sal	Eat	37	5	42	88
	No Eat	11	9	20	55
Cole Slaw	Eat	27	7	34	79
	No Eat	17	9	26	65

n=66

7.9)

	non	moder	heavy
Hyperten	21	36	30
No Hyperten	28	26	19

7.11)

♂	♀	f
3	0	7
2	1	35
1	2	3
0	3	2

Chi Goodness of fit

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7.16)

Num 6	Obs f
5	6
4	8
3	12
2	20
1	39
0	35

120 rolls of 5 dice

$p(x) = 1/6$

7.21)

	Papaveretum	Saline
pain better	47	7
No Change/Wor	3	42

7.27)

	Streptomycin	Rest
lived	51	38
died	4	14
	55	52